



BRIEF REPORT

Leveraging Real-World Evidence to Define Severe RSV Lower Respiratory Tract Disease in Adults

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ABSTRACT

Introduction: As no standard case definitions for respiratory syncytial virus—associated lower respiratory tract disease (RSV-LRTD) in adults are available, this study analyzed definitions for severe RSV-LRTD from previously published data in hospital and community cohorts of adults with RSV-associated symptoms.

Methods: The frequency, sensitivity, and specificity of acute respiratory disease symptoms

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among hospitalized and community cohorts of adults with RSV were analyzed. RSV-LRTD signs/symptoms assessed included shortness of breath (dyspnea), cough and/or fever, wheezing/rales/rhonchi (abnormal lung sounds by auscultation), sputum production, tachypnea, hypoxemia, and pleuritic chest pain.

Results: Dyspnea and tachypnea provided the best differentiation between hospitalized and community RSV-positive cases. The severe RSV-LRTD case definition yielding one of the highest and best-balanced sensitivity and specificity was dyspnea paired with either abnormal lung sounds by auscultation, hypoxemia, tachypnea, cough and/or fever, sputum, or chest pain.

Conclusions: Dyspnea alone, and in combination with certain other lower respiratory tract disease signs/symptoms, was a leading symptomatic indicator for severe RSV outcomes. These results contribute to the harmonization of case definitions for RSV disease.

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Key Summary Points

Respiratory syncytial virus (RSV) places a substantial disease burden on individuals of all ages, yet there is a lack of specific agreed-upon signs, symptoms, or standardized case definitions to differentiate RSV-acute respiratory disease (ARD), RSV-lower respiratory tract disease (RSV-LRTD), or severe RSV-LRTD in adults.

This study analyzed definitions of severe RSV-LRTD from previously published data in hospital and community cohorts of adults with RSV-associated symptoms that could be leveraged for the analysis of adult RSV vaccine efficacy trials and other clinical or observational studies.

Shortness of breath (dyspnea) was identified as a prominent symptom associated with severe outcomes in RSV cases.

This study also revealed that specific combinations of symptoms, such as dyspnea with wheezing/rales/rhonchi (abnormal lung sounds by auscultation), hypoxemia, tachypnea, cough and/or fever, sputum, or chest pain, offer the most effective differentiation identification of severe RSV disease.

These findings contribute to the standardization of RSV-LRTD case definitions in adults and highlight the importance of recognizing and monitoring dyspnea in RSV cases.

INTRODUCTION

Respiratory syncytial virus (RSV) affects people of all ages; older adults and individuals with certain underlying conditions are at higher risk of developing complications from RSV disease [1–3]. RSV typically causes mild upper respiratory tract symptoms [3, 4], but can progress to

severe lower respiratory tract disease (LRTD) in older and at-risk adults, potentially leading to pneumonia, hospitalization, and death [3, 4]. In 2019, RSV caused an estimated 5.2 million cases, 466,000 hospitalizations, and 33,000 in-hospital deaths among high-income countries in adults aged ≥ 60 years [5].

Laboratory testing is necessary to confirm an RSV diagnosis, as RSV is not easily clinically distinguishable from other respiratory pathogens [2]. In addition, there are no specific agreed-upon signs and symptoms to distinguish RSV-acute respiratory disease (ARD) from RSV-LRTD, or severe RSV-LRTD. Consequently, no standard case definitions exist for these outcomes in adults [2, 5, 6]. This, in addition to an overall lower sensitivity of rapid diagnostic tests and infrequent testing in adults [5, 7], may contribute to an underestimation of RSV-associated disease and present challenges when comparing results across vaccine and therapeutic studies.

In this study we performed a new analysis on previously collected data from a landmark observational cohort study (Falsey et al. [8]) to investigate definitions of severe RSV-LRTD that could be leveraged for the analyses of adult RSV vaccine efficacy trials.

METHODS

Population

The study retrospectively analyzed data from a cohort study of adults aged ≥ 18 years (median age, 73 years; Interquartile range, 18 years) who were evaluated for signs/symptoms of ARD and tested for RSV by reverse transcription-polymerase chain reaction in Rochester, New York [8]. The population analyzed included two cohorts: (1) individuals hospitalized with ≥ 1 ARD sign/symptom or an acute cardiopulmonary illness (hospitalized cohort), and (2) individuals enrolled and followed prospectively for ARD in the community (including those who did and did not seek medical care for ARD; community cohort). Additional population selection criteria are shown in the Supplemental Methods.

Probability of Individual or Paired ARD Signs/Symptoms

This analysis assessed the probability with which symptoms or combinations of symptoms were associated with positive versus negative RSV diagnoses in hospitalized versus community cohorts (e.g., proxies for severe vs less severe RSV disease), and in medically attended versus non-medically attended individuals (approximating moderate vs mild RSV disease). Symptoms were assessed as either new or worsening from baseline. A z-test was used to infer statistical differences between the frequencies (additional details are in the Supplemental Methods).

Assessment of Severe RSV-LRTD Case Definitions

Informed by the frequencies of signs/symptoms observed across a spectrum of RSV disease, we evaluated the performance of proposed severe RSV-LRTD case definitions in predicting hospitalized versus community cases, as well as medically attended versus non-medically attended community cases, by applying them to the dataset.

Sensitivity and Specificity Analyses

Sensitivity and specificity of case definitions were calculated to accurately identify severe RSV-LRTD cases (described in the Supplemental Methods). A machine learning approach was also used to predict severe RSV-LRTD among RSV-positive patients. Logistic regression with ridge regularization and Explainable Boosting Machine [9] models were trained using various input variables (Supplemental Table 1).

Ethics

This study involved secondary use of de-identified data that had previously received Institutional Review Board approval. As such,

additional ethics approval was not required, and informed consent was not applicable.

RESULTS

Participant Characteristics

The data retrospectively analyzed from the original study by Falsey et al. [8] included 999 symptomatic individuals, with 163 confirmed cases of RSV (RSV-positive) and 836 without RSV (RSV-negative; Fig. 1), as confirmed by reverse transcription-polymerase chain reaction. Among the 836 RSV-negative participants, 156 (18.7%) had evidence of another viral respiratory pathogen (48.1% had influenza A [$n=75$] and 23.7% had human metapneumovirus [$n=37$]). Both RSV-positive and RSV-negative groups were similar with respect to age and the proportion of participants reporting underlying chronic obstructive pulmonary disease, congestive heart failure, diabetes mellitus, renal failure, average baseline respiratory rate, and oxygen saturation (Supplemental Table 2). Among the 22 RSV-positive participants in the community cohort who sought medical care, 19 received care in the outpatient setting, 2 received care in both the emergency department and hospital setting, and 1 received care in both the outpatient and hospital settings. As these individuals were not initially hospitalized, they were included in the community cohort.

Comparison of Signs/Symptoms

RSV-Positive Versus RSV-Negative Cases

Overall, cough (90%), nasal congestion (79%), sputum production (72%), and abnormal lung sounds by auscultation (71%) were the most common signs/symptoms among RSV-positive participants ($n=163$). Compared with RSV-negative participants ($n=836$), RSV-positive participants more frequently reported wheezing or sputum production or had rhonchi on examination (Supplemental Fig. 1A).

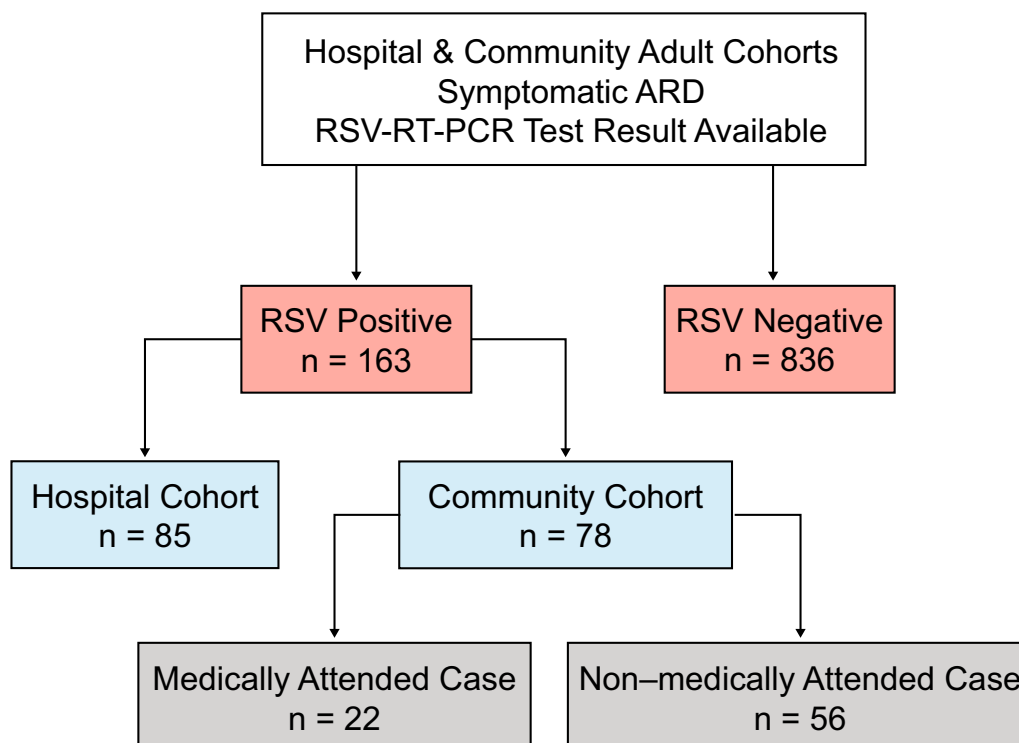


Fig. 1 Participants from Falsey et al. 2005 [8] were reanalyzed based on RSV RT-PCR results, whether they were identified in hospital or community settings, and within the community setting, whether participants went on to

seek medical attendance in the outpatient clinic, emergency department, or hospital. *ARD* acute respiratory disease, *RSV* respiratory syncytial virus, *RT-PCR* reverse transcription-polymerase chain reaction

RSV-Positive Cases: Hospitalized Cohort Versus Community Cohort

Shortness of breath (dyspnea; 98%), abnormal lung sounds by auscultation (98%), cough (94%), and tachypnea (91%) were the most common signs/symptoms among hospitalized RSV-positive patients ($n=85$). A greater percentage of hospitalized RSV-positive patients reported dyspnea, hypoxemia, tachypnea, rales, rhonchi, fever, or wheezing compared with those in the community cohort ($n=78$; Supplemental Fig. 1B).

Community Cohort RSV-Positive Cases: Medically Attended Versus Non-Medically Attended

For medically attended RSV-positive participants in the community cohort (i.e., ≥ 1 visit to the outpatient, emergency department, or

inpatient setting; $n=22$), cough (100%), sputum production (86%), and nasal congestion (82%) were the most reported ARD symptoms. A greater percentage of medically attended RSV-positive participants in the community cohort had sputum production, hypoxemia, or cough than non-medically attended RSV-positive participants in the community cohort. Dyspnea was observed in 41% of medically attended RSV-positive participants compared with 23% of non-medically attended RSV-positive participants ($n=56$; Supplemental Fig. 1C).

Sensitivity and Specificity Analyses of Severe RSV-LRTD Case Definitions

The highest sensitivity (100%) was observed with any combination of 2 of the following signs/symptoms: dyspnea, abnormal lung sounds by auscultation, fever and/or cough, hypoxemia, or tachypnea (Fig. 2; Supplemental

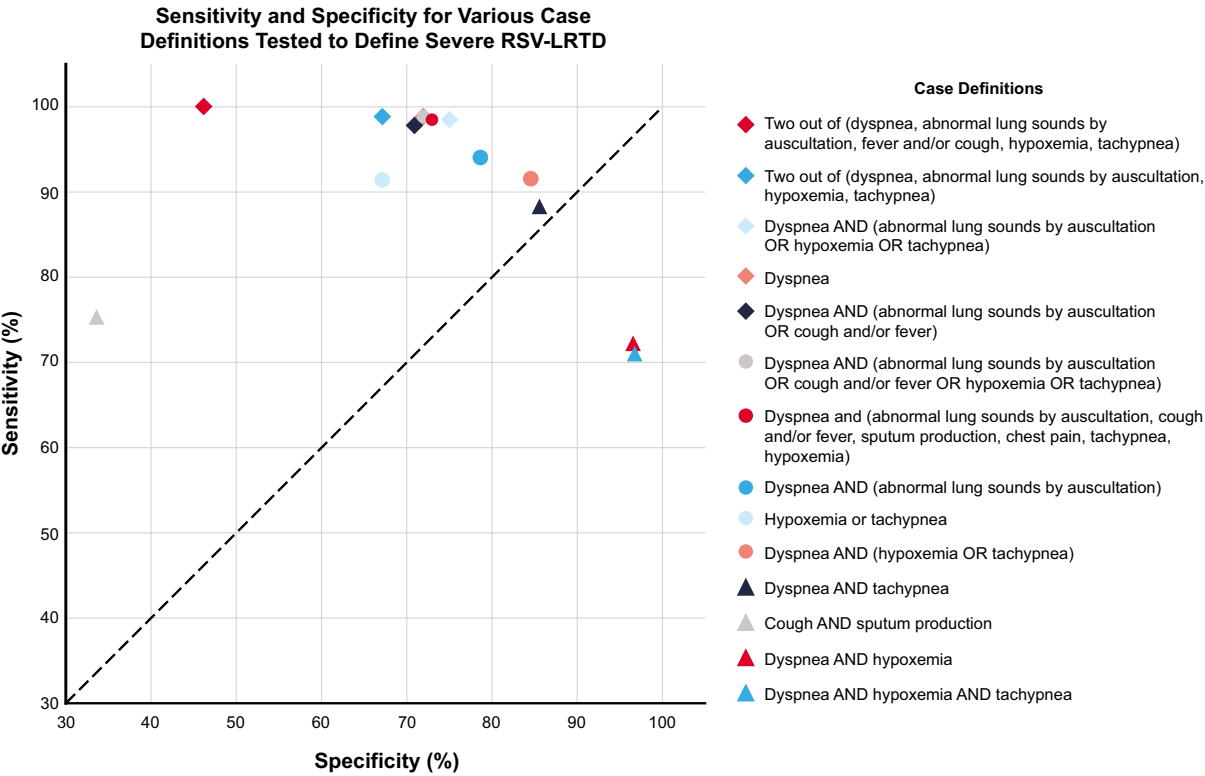


Fig. 2 Assessment of sensitivity and specificity across different case definitions for severe RSV-LRTD classification. Abnormal lung sounds by auscultation refers to wheeze/

rales/rhonchi. *LRTD* lower respiratory tract disease, *RSV* respiratory syncytial virus

Table 3); however, this was at the expense of specificity (46%). The definitions demonstrating the best balance between sensitivity (98%) and specificity (72%–76%) were those that included dyspnea alone, or either of the following variations: 1) dyspnea AND abnormal lung sounds by auscultation OR hypoxemia OR tachypnea; 2) dyspnea AND abnormal lung sounds by auscultation OR cough and/or fever; 3) dyspnea AND abnormal lung sounds by auscultation OR cough and/or fever OR hypoxemia OR tachypnea; 4) dyspnea AND abnormal lung sounds by auscultation, cough and/or fever, sputum production, chest pain, tachypnea, hypoxemia. Cough and sputum production were not specific for hospitalization compared with other definitions that may or may not have included cough. Probability of paired ARD signs/symptoms is discussed in the Supplemental Results.

Sensitivity and Specificity Analyses of Severe RSV-LRTD Using a Machine Learning Approach

A machine learning approach to predict severe RSV-LRTD (e.g., hospitalization from RSV) had a sensitivity of 1.0 and specificity of 0.80 for logistic regression, and a sensitivity of 0.94 and specificity of 0.86 for Explainable Boosting Machine (Supplemental Table 4; additional information available in the Supplemental Results). Consistent with results from the non-machine learning approach, the variables with the largest weights (i.e., ranked as most important in predicting hospitalization) were dyspnea (Supplemental Fig. 2), followed by hypoxemia, tachypnea, and fever. Other variables that were found to impact risk of hospitalization included age, pulse, rales, and combination of dyspnea and body temperature.

DISCUSSION

This is the first study to our knowledge to reanalyze data from a prospective observational cohort to develop case definitions for severe RSV-LRTD in adults. Dyspnea as a symptom alone had a notably high sensitivity and specificity for predicting severe outcomes (i.e., hospitalization). Across the spectrum of increasingly severe RSV disease, the frequency of dyspnea increased from 23% in the RSV non-medically attended community cohort to 41% in the RSV medically attended community cohort to 98% in the hospitalized cohort. Given the high sensitivity of dyspnea alone for predicting RSV-LRTD-associated hospitalizations in our analysis, this symptom should be considered as a leading symptomatic indicator of LRTD and severe RSV disease in future studies. As other studies have suggested, the analyses showed that no individual or combination of signs/symptoms was highly specific in distinguishing between RSV-positive and RSV-negative individuals, although there were general trends [2]. Hence, laboratory testing combined with a comprehensive list of LRTD signs/symptoms are critical features of any RSV-LRTD case definition.

This study had several limitations, including a relatively small sample size in certain groups, limited geographic scope, and a population that included a high proportion of individuals with COPD and self-reported smokers, thus limiting its generalizability. Study strengths included prospective and systematic collection of signs and symptoms across both hospital and community cohorts, and the collection of data over 4 RSV seasons.

This study demonstrates how data from a well-characterized observational cohort can inform case definitions in a clinical trial. Additionally, this study provides information regarding which signs and symptoms of RSV disease are related to healthcare-seeking behavior and hospitalizations in adult cohorts, where markers for severe RSV disease are not standardized. As the study population included adults of all ages, these data can be considered in developing case definitions for

severe RSV-LRTD across the adult lifespan, including older adults and younger high-risk populations.

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Author Contributions. Catherine A. Panozzo designed the concept. Catherine A. Panozzo, Edward E. Walsh, Zhen Yang, Eleanor Wilson, Jaya Goswami, Sonia K. Stoszek, Adrianna Loback, Tony Ng, Beverly M. Francis, Alana K. Simorellis, Linwei Li, Evan J. Anderson, and Ann R. Falsey analyzed or interpreted the data. All authors provided writing, review, or intellectual contributions and approved the final draft.

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Data Availability. The authors declare that the data supporting the findings of this study are available within this article.

Declarations

Conflict of interest. Catherine A. Panozzo, Zhen Yang, Eleanor Wilson, Jaya Goswami, Sonia K. Stoszek, Beverly M. Francis, Alana K. Simorellis, Wenmei Huang, Linwei Li, Evan J. Anderson, Rebecca Vislay-Wade, and Zhe Zheng are employees of Moderna, Inc., and hold stock/stock options in the company. Adrianna Loback, Tony Ng, Allison August, and Grace Chen. were employees of Moderna, Inc., at the time of study conduct. Ann R. Falsey has received research grants from Pfizer, Janssen, Merck, CyanVac, BioFire Diagnostics, and Vaccine Company; has consulted for Sanofi Pasteur and Arrowhead Pharmaceuticals; and has received payment from Novavax DSMB. Edward E. Walsh has received research grants from Pfizer and Merck and has served on the advisory board for GSK and Moderna, Inc.

Ethical Approval. This study involved secondary use of de-identified data that had previously received Institutional Review Board approval. As such, additional ethics approval was not required, and informed consent was not applicable.

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